CUSTOMER CHURN PREDICTION AND RETENTION STRATEGY

Individual Task - Customer Churn Prediction & Retention Analysis.

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1. Executive Summary

This project develops an end-to-end machine learning solution to predict customer churn and support data-driven retention decisions for a subscription service provider. The primary business challenge is identifying customers most likely to leave when the company can contact only 20% of its customer base.

The analysis uses a dataset of 1,200 customers containing demographic, subscription, billing, satisfaction, complaint, payment and service-usage information. Data auditing, exploratory analysis, feature engineering, preprocessing, model development, hyperparameter optimization, probability calibration and explainability were performed.

Three classification algorithms were evaluated: Logistic Regression, Random Forest and Gradient Boosting. Following cross-validation and hyperparameter tuning, a sigmoid-calibrated Random Forest was selected as the final research model.

Key Results

- **Test Accuracy:** 87.50%
- **Test ROC-AUC:** 0.9157
- **Test PR-AUC:** 0.9851
- **Churn Recall:** 97.58%
- **Top-20% Precision:** 100%
- **Top-20% Lift over Random:** 1.1594×

On the independent test dataset, the model identified all 48 actual churners among the 48 highest-risk customers selected for contact. Random selection would identify approximately 41.4 churners on average.

A hypothetical retention simulation using development data estimated an incremental net benefit of **726 currency units** compared with random selection, assuming a 25% successful retention rate and a retained-customer value of 120 currency units.

2. Business Objective

The objective is to develop an explainable customer churn prediction system that enables the company to prioritize retention activities within a limited contact budget.

Rather than relying solely on classification accuracy, the project emphasizes risk ranking, probability reliability, model explainability and measurable business impact.

Management Recommendation: The calibrated Random Forest demonstrates potential for prioritizing retention outreach. However, the reliability and timing of monthly usage data must

be verified before operational deployment. Financial benefits remain hypothetical until retention interventions are validated experimentally.

3. Dataset Overview

The analysis uses the Customer Churn Advanced Dataset, containing **1,200 customer records and 20 variables**. The dataset includes demographic characteristics, subscription information, billing details, service usage, customer satisfaction, complaints, and payment behavior.

The target variable, churn, represents whether a customer has left the subscription service (1 = Churned, 0 = Retained).

3.1 Dataset Summary

Characteristic	**Value**
Total Customers	1,200
Total Variables	20
Churned Customers	1,036 (86.33%)
Retained Customers	164 (13.67%)
Missing Values	138 cells
Duplicate Records	0
Training Dataset	960 customers (80%)
Testing Dataset	240 customers (20%)

The dataset exhibits substantial class imbalance, with approximately 86% of customers belonging to the churned class. Consequently, model evaluation must consider balanced accuracy, ROC-AUC, average precision, and business-oriented risk ranking rather than accuracy alone.

4. Data Quality Assessment

A systematic data audit was performed to identify missing values, duplicate records, invalid values, and unusual observations.

4.1 Missing Value Analysis

Feature	Missing Values	Percentage
satisfaction_score	48	4.00%
monthly_usage_gb	36	3.00%
monthly_charges	30	2.50%
total_charges	24	2.00%
Total	**138**	

Missing values were preserved in the cleaned dataset. Numerical imputation was performed within the machine learning pipeline using training-derived medians to prevent data leakage.

4.2 Data Quality Findings

No duplicate records or duplicate customer identifiers were detected.

However, 147 customers had positive monthly usage despite their internet service being recorded as "No". Additionally, five records exhibited substantial differences between cumulative charges and simplified billing estimates.

These observations were flagged for investigation rather than automatically deleted because they may represent legitimate business scenarios or differences in data definitions.

4.3 Data Preparation

The cleaning process standardized categorical text, validated numerical fields, and preserved potentially legitimate outliers.

A reproducible Scikit-learn preprocessing pipeline was developed using Pipeline and ColumnTransformer. Numerical variables were processed using median imputation and standard scaling, while categorical variables were processed using most-frequent imputation and one-hot encoding.

Customer identifiers were excluded from model training. The potentially leakage-prone total_charges feature was also excluded from the baseline predictive feature set.

Outcome: A cleaned dataset containing all 1,200 customer records was exported, with missing values retained for leakage-safe pipeline imputation.

5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted using 960 training customers to investigate patterns associated with customer churn. Nine initial visualizations and three advanced analyses were developed, covering subscription contracts, tenure, billing, satisfaction, complaints, payment behavior and service usage.

5.1 Contract Type and Customer Churn

Customers with month-to-month contracts exhibited the highest churn rate at **91.1%**, compared with 80.0% for one-year contracts and 79.1% for two-year contracts.

This suggests that customers with flexible subscription arrangements may represent an important group for retention monitoring.

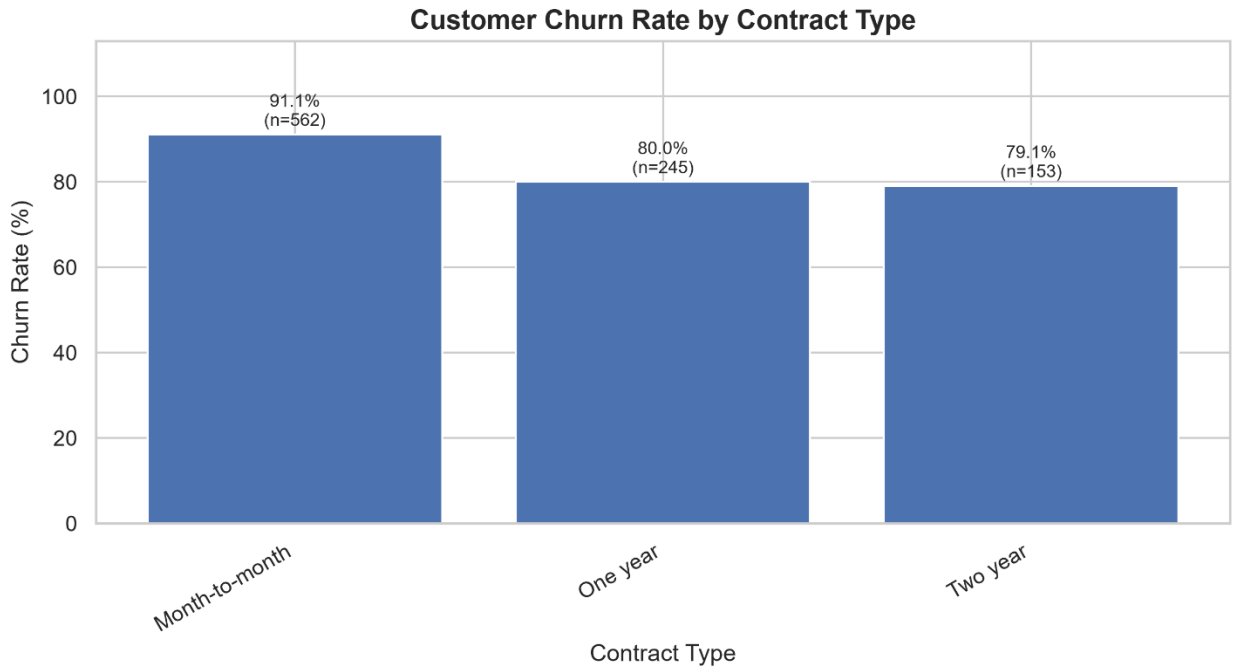


Figure 1: Customer Churn Rate by Contract Type

5.2 Customer Satisfaction and Churn

Customer satisfaction demonstrated a negative association with churn. Customers with satisfaction scores of 1–4 exhibited a churn rate of **92.3%**, compared with 81.2% among customers scoring above 8.

Lower satisfaction may therefore be a useful indicator when prioritizing customers for further investigation.



Figure 2: Customer Churn Rate by Satisfaction Score

5.3 Monthly Usage and Churn

Monthly internet usage demonstrated the strongest observed relationship with churn. Customers in the lowest usage quartile exhibited a churn rate of **62.6%**, increasing to approximately **99.1%** among customers in the highest usage quartile.

This substantial difference suggests that usage information contains valuable predictive signals.

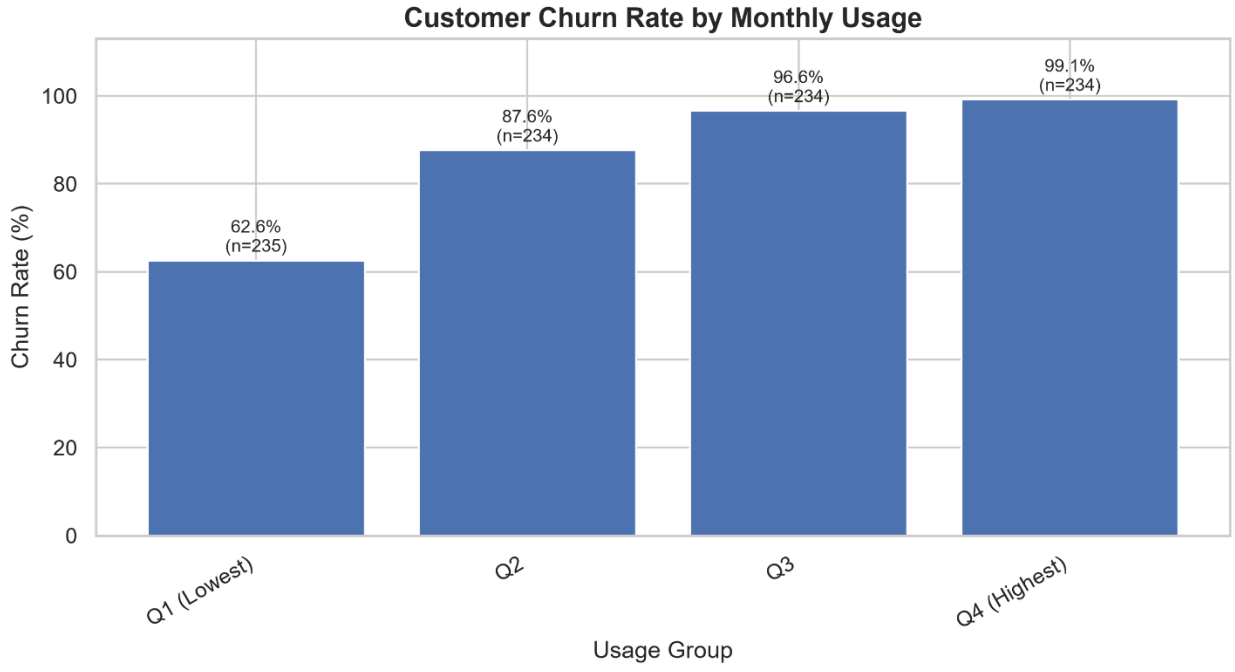


Figure 3: Customer Churn Rate by Monthly Usage

5.4 Key Business Insights

The exploratory analysis identifies three major churn-risk patterns: month-to-month contracts, lower satisfaction scores and higher recorded monthly usage.

These findings support developing retention strategies informed by customer behavior. However, the relationships are predictive associations rather than evidence of causation.

Additionally, inconsistencies between internet-service categories and reported usage indicate that usage information requires further verification before operational deployment.

6. Feature Engineering and Preprocessing

Eight engineered features were developed to capture additional customer behavior patterns identified during exploratory analysis.

6.1 Engineered Features

Feature	**Business Justification**
Tenure Years	Represents customer relationship duration
Payment Risk Flag	Identifies customers with multiple late payments
Complaint Intensity	Measures complaints relative to observation duration
Total Service Issues	Combines complaints and late-payment counts
Monthly Charge per GB	Represents charges relative to service usage
Contract-Payment Interaction	Captures late payments for month-to-month customers
Low Satisfaction Flag	Identifies customers with satisfaction scores of 4 or below
Charge-Satisfaction Interaction	Combines billing level with dissatisfaction

The engineered features increased the predictive input variables from 17 to 25.

A reproducible Scikit-learn pipeline was implemented using Pipeline and ColumnTransformer. Numerical missing values were handled through median imputation, followed by standard scaling. Categorical variables were processed through most-frequent imputation and one-hot encoding.

To prevent data leakage, all learned preprocessing transformations were fitted within training folds. Customer identifiers and cumulative total charges were excluded from the baseline predictive feature set.

7. Machine Learning Model Development

Three supervised classification algorithms were implemented: Logistic Regression, Random Forest and Gradient Boosting.

Logistic Regression provided an interpretable linear baseline, while Random Forest and Gradient Boosting were introduced to capture nonlinear relationships and feature interactions.

An 80:20 stratified train-test split was used, with 960 training customers and 240 customers reserved for independent testing. Model development used stratified five-fold cross-validation.

7.1 Baseline Model Comparison

Model	**ROC-AUC**	**PR-AUC**	**Balanced Accuracy**
Logistic Regression	0.8683	0.9722	0.6756
Random Forest	0.8509	0.9658	0.5999
Gradient Boosting	**0.8745**	**0.9734**	0.6672

Gradient Boosting achieved the highest baseline ROC-AUC and PR-AUC, while Logistic Regression achieved slightly higher balanced accuracy.

7.2 Hyperparameter Optimization

RandomizedSearchCV was applied to all three models using stratified five-fold cross-validation.

The optimized Random Forest achieved the highest cross-validated ROC-AUC of **0.8811**, compared with 0.8697 for Logistic Regression and 0.8759 for Gradient Boosting.

The tuned Random Forest used 300 trees, maximum depth of 10, minimum leaf size of 4 and balanced-subset class weighting.

These results supported selecting Random Forest for further probability calibration and business-oriented evaluation.

8. Final Model Evaluation

The sigmoid-calibrated Random Forest was selected as the final research model based on cross-validation, hyperparameter optimization, probability calibration and business-oriented evaluation.

The model was trained using 960 customers and independently evaluated on 240 previously unseen test customers.

8.1 Independent Test Performance

Evaluation Metric	**Test Result**
Accuracy	87.50%
Balanced Accuracy	60.91%
Precision (Churn)	88.99%
Recall (Churn)	97.58%
F1 Score (Churn)	93.09%
ROC-AUC	**0.9157**
PR-AUC (Average Precision)	**0.9851**
Brier Score	0.0805
Log Loss	0.2539

The final model demonstrated strong discrimination, achieving ROC-AUC of 0.9157 and PR-AUC of 0.9851.

At a classification threshold of 0.50, the model identified approximately 97.6% of actual churners. However, retained-customer recall was only approximately 24%, indicating difficulty distinguishing customers who remain.

Overall accuracy exceeded the majority-class baseline of 86.25% by only 1.25 percentage points, demonstrating why accuracy alone is insufficient.

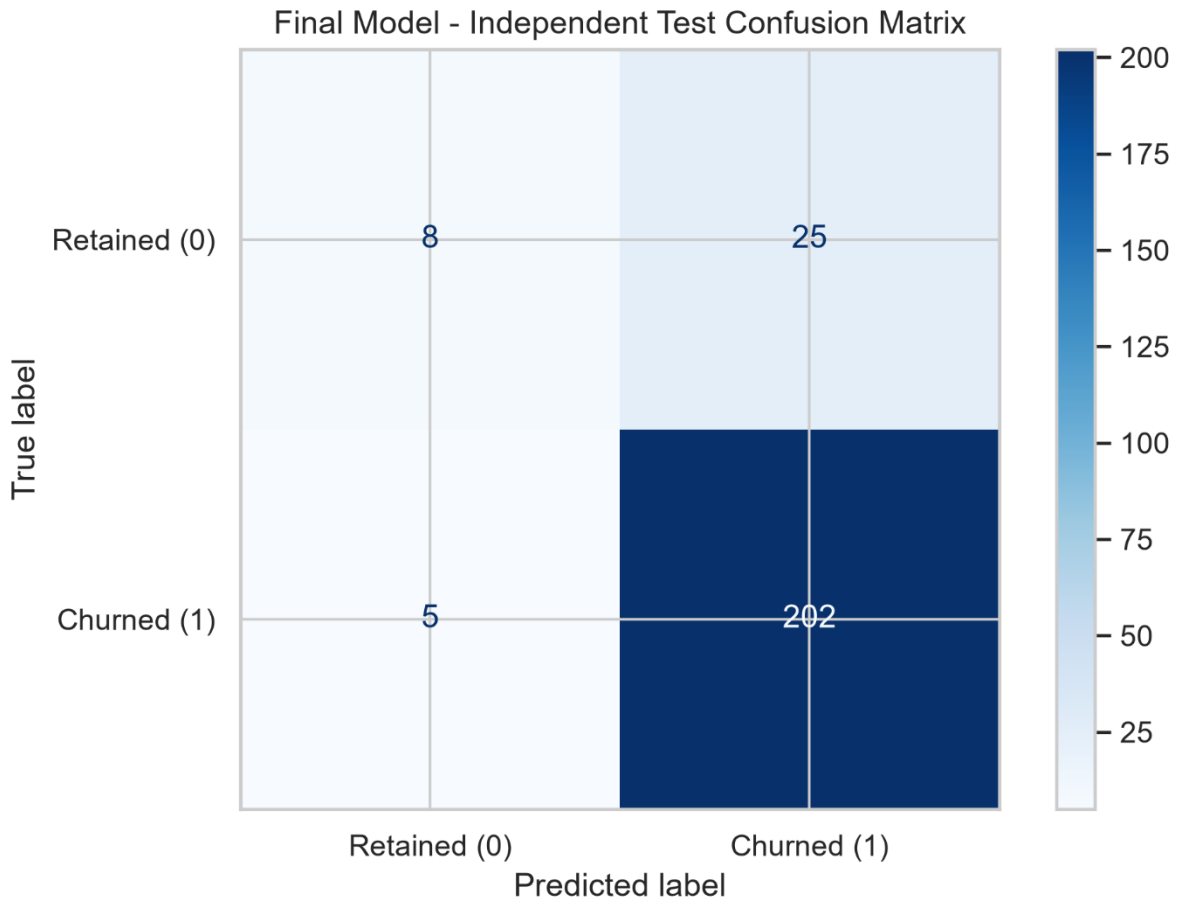


Figure 4: Final Test Confusion Matrix

9. Probability Calibration

Probability calibration was investigated to assess whether predicted churn probabilities accurately reflected observed churn frequencies.

Sigmoid calibration was applied to the tuned Random Forest using cross-validation.

9.1 Calibration Comparison

Metric	Uncalibrated	Calibrated
Brier Score	0.0917	**0.0842**
Log Loss	0.3249	**0.2783**
ROC-AUC	**0.8778**	0.8737

PR-AUC	**0.9754**	0.9710

These development-stage results show that calibration reduced Brier Score by approximately 8.2% and Log Loss by 14.3%, while causing minor reductions in ranking performance.

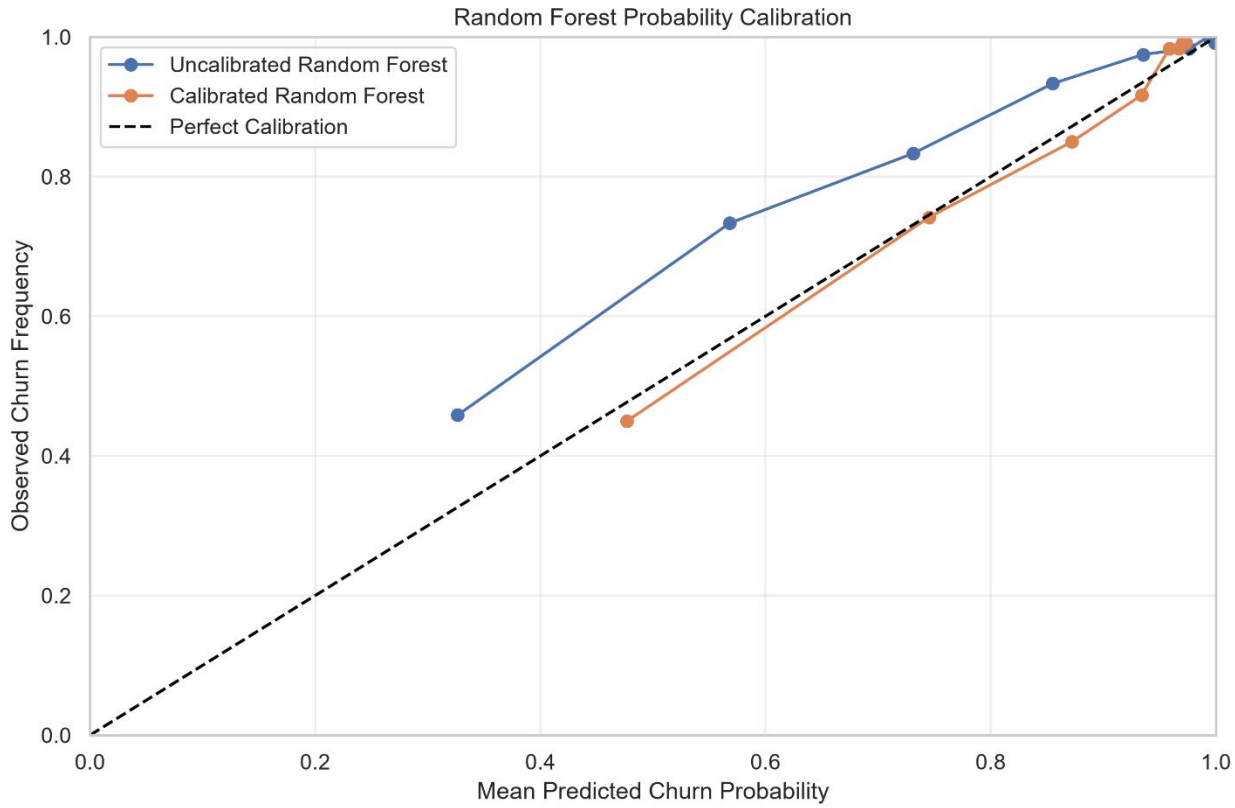


Figure 5: Random Forest Probability Calibration Curve

The results support using calibrated probabilities for research and retention prioritization, subject to the dataset's limitations. Operational contact decisions use the highest-ranked 20% of customers rather than an arbitrary probability threshold.

10. Model Explainability

Explainability analysis was conducted to understand the predictions generated by the calibrated Random Forest. Global permutation importance and individual customer-level sensitivity analysis were used to investigate model behavior.

10.1 Global Feature Importance

Feature	Permutation Importance
Monthly Internet Usage	**0.3454**
Contract Type	0.0382
Monthly Charges	0.0170
Satisfaction Score	0.0059
Customer Tenure	0.0058

Monthly internet usage was the strongest predictor. Shuffling this feature reduced validation ROC-AUC by approximately 0.3454, indicating that the model relies heavily on usage information.

Contract type and monthly charges were the next most influential variables.

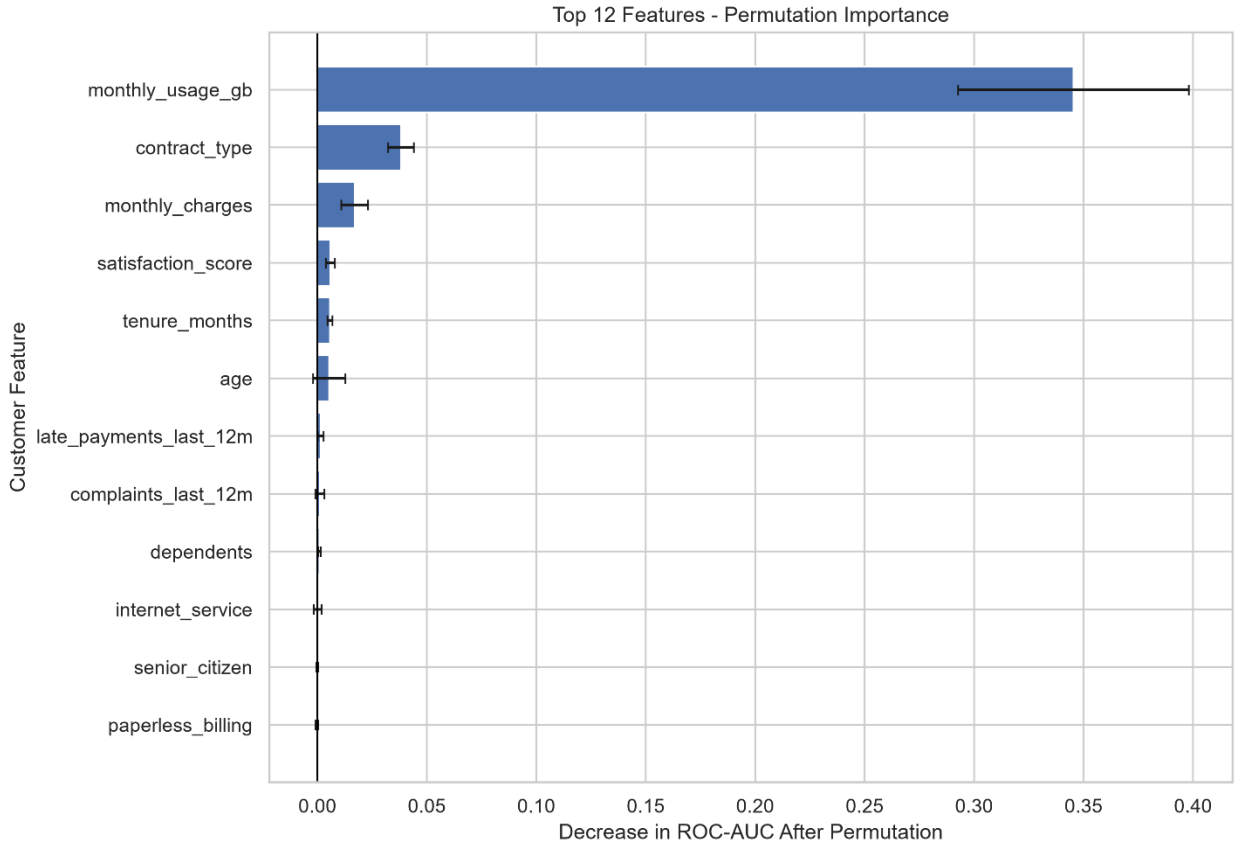


Figure 6: Global Permutation Feature Importance

10.2 Individual Customer Explanations

Three customers were selected from an internal validation subset to examine individual model predictions.

Customer	Predicted Churn Risk	Key Finding
Highest Risk	96.59%	Replacing usage of 113.6 GB with 65.6 GB reduced predicted risk to 84.58%.
Median Risk	93.91%	Region, late payments and charges produced comparatively small prediction changes.
Lowest Risk	27.15%	Replacing usage of 11.3 GB with 65.6 GB increased predicted risk to 78.74%.

These explanations demonstrate that the model's predictions are particularly sensitive to monthly usage. The results describe predictive associations and model sensitivity, not causal relationships.

11. Model Robustness Analysis

A feature-ablation experiment was conducted to assess model performance when monthly usage and the derived charge-to-usage ratio were excluded.

11.1 Robustness Comparison

Metric	Full Model	Without Usage
Training ROC-AUC	0.9961	0.9995
Validation ROC-AUC	**0.8811**	0.6192
PR-AUC	**0.9770**	0.9071
Balanced Accuracy	**0.7193**	0.5180
Generalization Gap	0.1151	0.3803

Removing monthly usage reduced validation ROC-AUC by **0.2619**, demonstrating substantial dependence on this feature.

Both models exhibited training-validation performance gaps, particularly the reduced-feature model.

Key Limitation: Earlier data auditing identified inconsistencies between internet-service status and usage values. The dataset also lacks information confirming whether usage was measured before churn occurred.

Therefore, the model should not be deployed operationally until the definition, timing and reliability of monthly usage have been verified.

12. Customer Retention Intervention Simulation

The business objective is to identify customers at high risk of churn while respecting a contact capacity limited to 20% of the customer base.

Customers were ranked using predicted churn probabilities from the calibrated Random Forest. The highest-risk 20% were selected for potential retention interventions.

12.1 Independent Test Retention Results

Performance Indicator	**Result**
Total Test Customers	240
Contact Capacity (20%)	48
Actual Churners	207
Churners Identified by Model	**48**
Random Selection (Expected)	41.4
Precision@20%	**100%**
Recall@20%	23.19%
Lift Over Random	**1.1594**×

The model successfully identified 48 actual churners among the 48 highest-risk test customers.

Compared with random selection, the model identified approximately 6.6 additional churners in expectation, representing a 15.94% improvement in targeting efficiency.

Although Recall@20% was 23.19%, this was close to the theoretical maximum achievable under the 20% contact constraint.

13. Financial Cost-Benefit Analysis

A hypothetical financial simulation was conducted using cross-validated development predictions to estimate the potential value of model-based retention targeting.

13.1 Business Assumptions

Assumption	Value
Customers Contacted	192
Cost per Contact	5 CU
Retained Customer Value	120 CU
Retention Success Rate	25%

CU represents hypothetical currency units.

13.2 Estimated Financial Outcomes

Metric	Random	Model-Based
Churners Identified	165.8	190
Expected Customers Retained	41.45	47.50
Gross Benefit (CU)	4,974	5,700
Contact Cost (CU)	960	960
Net Benefit (CU)	**4,014**	**4,740**

The model-based strategy generated an estimated incremental net benefit of **726 CU**, equivalent to an 18.1% increase over random targeting under the stated assumptions.

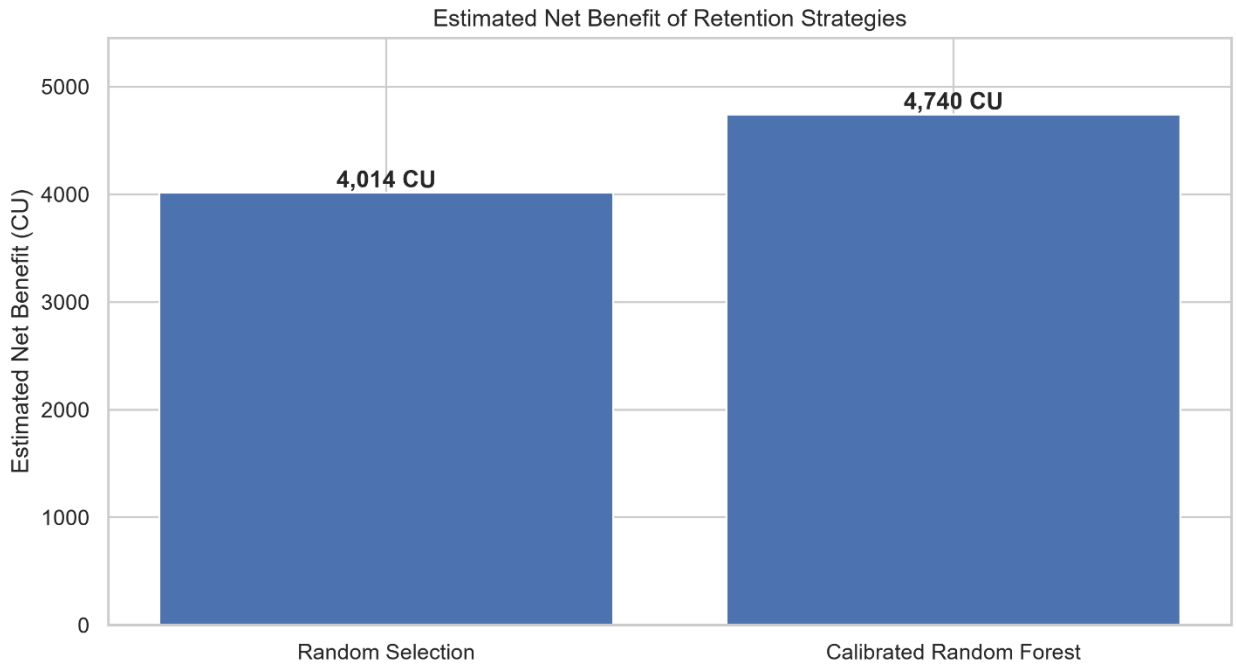


Figure 7: Estimated Financial Benefit of Retention Strategies

13.3 Sensitivity Analysis

Sensitivity analysis varied retention success rates between 10% and 50% and retained-customer values between 80 and 160 CU.

Estimated incremental benefits ranged from **193.60 CU to 1,936 CU** across the evaluated scenarios.

These findings suggest potential financial value from model-based customer targeting. However, the simulation assumes hypothetical intervention effectiveness and customer value.

Predicting churn does not guarantee that a customer can be successfully retained. Actual financial benefits must be validated through controlled retention campaigns.

14. Business Recommendations

Based on exploratory analysis, model explainability and retention simulation results, the following five recommendations are proposed for the subscription service provider.

14.1 Implement Risk-Based Customer Prioritization

Use the calibrated Random Forest to rank customers by predicted churn probability and prioritize the highest-risk 20% for retention outreach.

The model identified all 48 actual churners among the 48 highest-risk customers in the independent test dataset.

Recommendation: Introduce a controlled pilot campaign comparing model-based targeting with random targeting to evaluate actual retention effectiveness.

14.2 Improve Month-to-Month Customer Retention

Month-to-month customers demonstrated a churn rate of approximately 91.1%, higher than customers with longer subscription contracts.

Recommendation: Offer suitable loyalty benefits, flexible renewal incentives and personalized contract options to high-risk month-to-month customers.

14.3 Strengthen Customer Satisfaction and Service Support

Customers with lower satisfaction scores exhibited higher observed churn rates. Complaint and payment histories may also provide additional information about customer difficulties.

Recommendation: Prioritize proactive service follow-ups, complaint resolution and satisfaction surveys for customers showing multiple risk indicators.

14.4 Investigate High-Usage Customer Behavior

Monthly internet usage was the strongest predictive feature, with permutation importance of 0.3454.

However, high usage should not automatically be interpreted as a cause of churn.

Recommendation: Investigate whether high-usage customers experience service-quality problems, unsuitable pricing plans or unmet expectations. Validate usage-data definitions and measurement timing before using this feature operationally.

14.5 Establish Model Monitoring and Human Oversight

Recommendation: Regularly monitor prediction accuracy, ROC-AUC, calibration, customer-group performance and top-20% targeting effectiveness.

Retention teams should review recommendations before contacting customers, with appropriate data privacy protections and human oversight.

15. Project Limitations

The following limitations must be considered:

- **Class Imbalance:** Approximately 86% of customers belong to the churned class, making accuracy alone an insufficient performance measure.

- **Data Reliability:** Inconsistencies were identified between internet-service categories and recorded monthly usage.
- **Temporal Leakage Risk:** The dataset does not establish whether all predictive variables were measured before the churn outcome.
- **Model Dependence:** Removing monthly usage reduced validation ROC-AUC from 0.8811 to 0.6192.
- **Generalization:** The independent test set contained only 240 customers, and performance may differ on future customer populations.
- **Financial Assumptions:** The retention simulation uses hypothetical costs, customer values and intervention success rates.
- **Causality:** Predictive associations do not establish that changing customer characteristics will prevent churn.

These limitations indicate that the solution should be treated as a research prototype rather than a production-ready retention system.

16. Conclusion

This project successfully developed an end-to-end machine learning solution for customer churn prediction and retention analysis.

The final calibrated Random Forest achieved **0.9157 ROC-AUC**, **0.9851 PR-AUC** and **87.50% accuracy** on the independent test dataset.

Under the 20% contact-capacity constraint, the model identified 48 actual churners among 48 selected customers, outperforming random targeting.

A hypothetical development-data simulation estimated an incremental net benefit of **726 currency units** compared with random selection.

Overall, the project demonstrates how machine learning can support explainable, data-driven customer retention decisions.

Final Recommendation: Proceed with data validation and a controlled retention pilot before operational deployment. Evaluate actual intervention effectiveness, model reliability and ongoing business value before expanding the system.